Building AI Agents in Python

Using Github Copilot SDK

Copilot Dev Days | Trivandrum

whoami

Adarsh Divakaran

Python Developer Advocate @ SerpApi

Microsoft MVP in Python

AGENDA

01 INTRO

Intro to Copilot SDK
What are AI Agents

02 AGENT SETUP

Choosing an LLM
System Prompt
Copilot Sessions

03 DATA & TOOLS

Tools
MCP
Skills
Copilot Hooks & Permissions

04 FINALE

Demo Agent
Next Steps

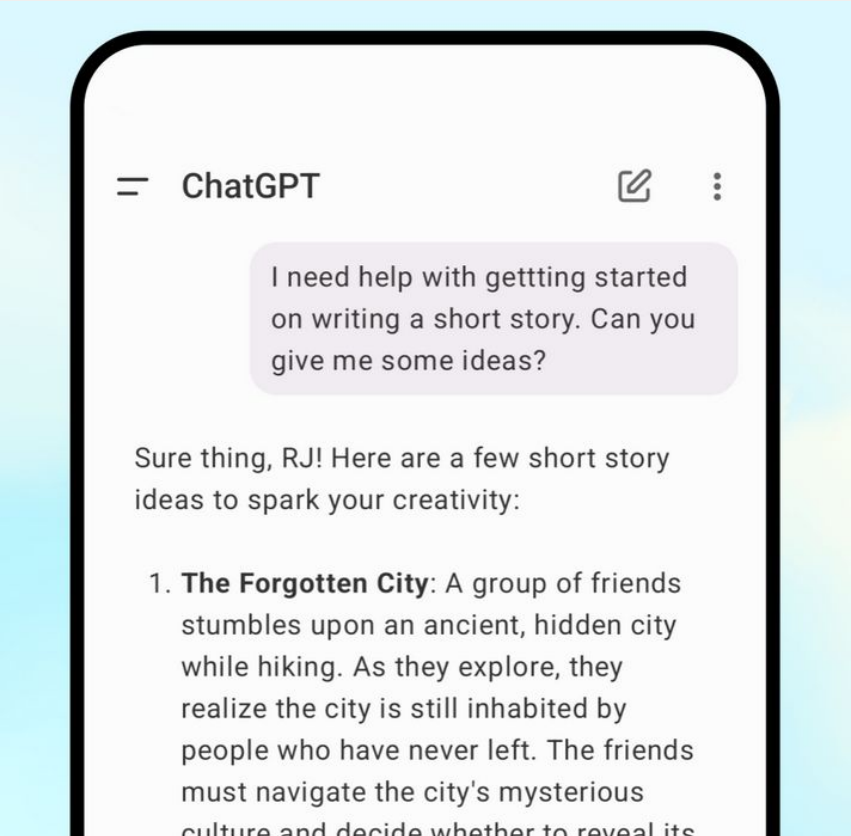
LLMs

- Trained an huge amounts of Data
- Good at a large number of tasks

Interacting with LLMs

- Most popular way: Chat

Fix writing, identify errors in code, summarize PDFs, etc.



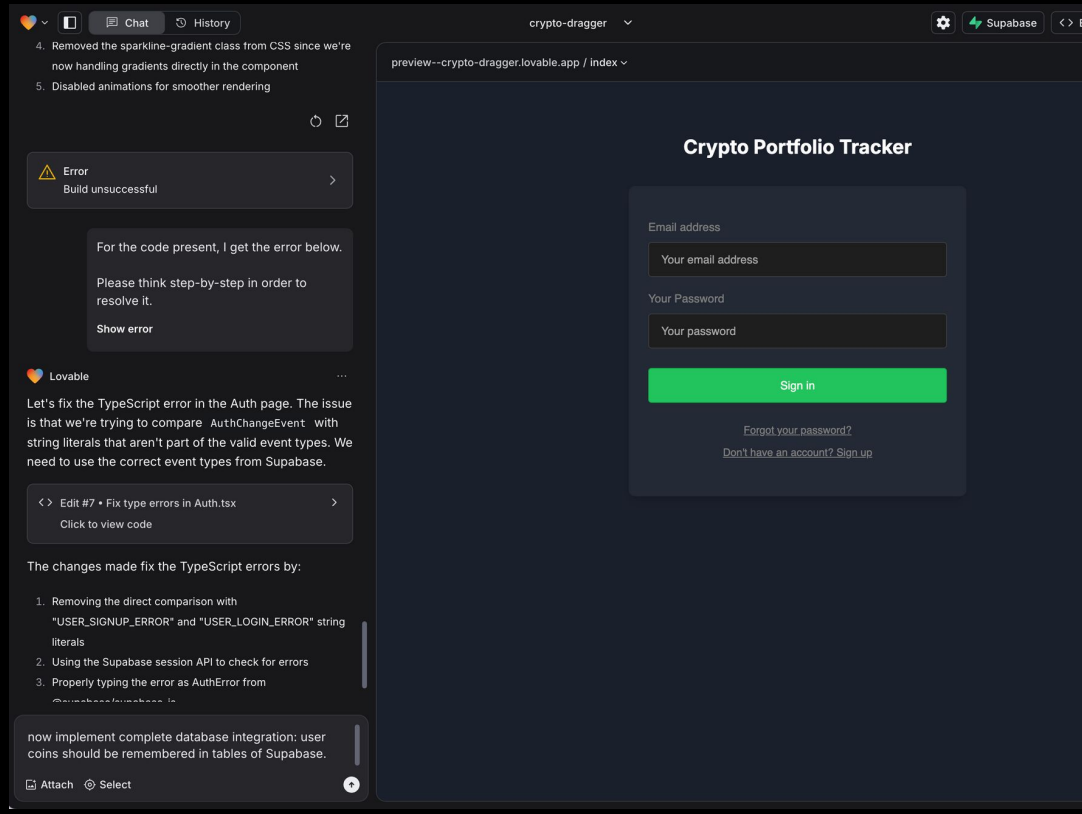
Agents

LLMs are capable to do more than just chat.

AI agent is an **AI system** that can **autonomously** plan and execute multi-step actions toward a **goal**.

Agents usually have specialized **tools**: Bash, Websearch, LSP

- Claude Code: Agentic coding system
- Lovable: Agentic Website builder



Chatbot vs Agent

A chatbot answers; an agent does work under rules set by the host application.

Chatbot flow: User message -> Model response

Agent flow: User goal -> App context -> Model -> Tool or question -> Result -> Final output

- the model is not the whole product
- the host application controls the loop, tools, permissions, and user experience

Choosing an LLM

- Benchmarks
- Costs
- Reasoning model
- Context Window

Choosing an LLM

Things to compare:

- Does the model follow instructions well? Can it call tools reliably?

- Is it fast enough for the UI?

- Do we need a hosted model or a local model?

Demo

Writing Helper App

Whats Next

- Memory: Persist information across sessions
- Agent Orchestration: Multiple Agents
- Evals: Testing model outputs
- Observability: Logs, tracing tools

References

- The Missing Playbook for AI Agents - Blog post by Kumar Anirudha
- AI Agent for beginners - Github repo by Microsoft
- A practical guide to building agents - Guide by OpenAI
- Building effective agents - Guide by Anthropic
- Patterns for Building AI Agents - Free Ebook by Mastra

Questions?

THANK YOU

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